Project Name: Green Lab Monitoring System

1. Overall Approach

To address the problem of **water, gas, and electricity wastage in college laboratories**, a **Centralized Smart Monitoring Architecture** has been adopted.

This solution replaces manual checking with an **automated sensor-driven workflow**.

The core strategy is **Real-Time Resource Tracking**.
By continuously collecting structured sensor data from lab resources, the system removes dependency on manual supervision.

The application provides distinct interfaces:

- **Faculty Dashboard:** focuses on live resource monitoring and issue alerts
- **Admin Dashboard:** focuses on analytics, monthly reports, and maintenance management
- **Maintenance Panel:** focuses on leak tickets and repair workflow

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2. System Architecture Overview

The system follows a **modular 3-Tier Architecture**.

Frontend (Presentation Layer)

- **Faculty View:** displays live water, gas, and electricity usage for each lab
- **Admin View:** shows lab-wise comparison, monthly reports, leaderboard, and waste analytics
- **Maintenance View:** displays unresolved alerts and generated repair tickets

Backend Logic (Business Layer)

- Acts as the **central controller** to process incoming sensor data
- Runs **Leak Detection Logic** to identify abnormal water or gas flow
- Executes **High Usage Detection Engine** for unusual electricity consumption
- Triggers **alert workflow and ticket generation**
- Handles dashboard APIs using REST architecture

Data Persistence Layer

Stores structured records of:

- sensor usage logs
- alert history
- maintenance complaints
- cost records
- monthly analytics reports

This ensures reliable reporting and trend analysis.

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3. Key Components

The system is composed of **four primary logical engines**.

Resource Monitoring Engine

Continuously collects:

- water usage
- gas flow
- electricity consumption

for each laboratory.

Smart Waste Detection System

Logic:

The backend compares current usage with predefined safe thresholds.

Action:

If abnormal flow or excessive consumption is detected:

- alert is triggered
- waste event is recorded
- admin is notified

Cost Tracking Engine

Calculates:

- lab-wise electricity cost
- monthly water usage cost

- gas refill estimates
- total waste reduction savings

This helps management optimize lab expenses.

Maintenance Ticket Generator

Whenever a leak or abnormal usage persists:

- a ticket ID is created
- assigned to maintenance staff
- tracked until resolution

This reduces delay in issue fixing.

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4. Data Flow Explanation

1. Data Collection

Sensors installed on:

- water pipes
- gas cylinders
- electricity meters

send live readings to backend services.

2. Validation & Detection

The backend checks:

- threshold crossing
- leak conditions
- sudden spikes
- devices left ON after lab hours

3. Storage

Validated data is stored with:

- timestamp
- lab ID

- resource type
- usage amount

4. Alerting

If waste is detected, the system generates:

- Leak Alert
- High Usage Alert
- Equipment Left ON Alert

5. Visualization

The dashboard updates instantly with:

- live usage
- comparison graphs
- green lab ranking
- pending issues

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5. Design Decisions

- **Web Dashboard:** selected for cross-device accessibility without installation
- **Real-Time Sensors:** used to eliminate dependency on manual lab checking
- **Threshold-Based Alerts:** implemented to instantly identify leaks and abnormal usage
- **Leaderboard UX:** added to encourage competition among labs for resource saving
- **Modular Backend:** chosen for scalability across multiple departments and buildings

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6. One-Line Presentation Summary

The Green Lab Monitoring System uses IoT sensors, backend intelligence, and real-time dashboards to reduce water, gas, and electricity wastage in laboratories while improving safety and maintenance efficiency.